

JARVIS Test Document

Artificial Intelligence and Machine Learning Basics

This document is designed as a sample PDF for testing the JARVIS document-question-answering system. JARVIS should read this document and answer questions using the information contained in it.

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computer science concerned with creating systems that can perform tasks that normally require human intelligence. These tasks can include understanding language, recognizing images, making predictions, solving problems, and supporting decisions.

2. What is Machine Learning?

Machine Learning (ML) is a branch of AI in which computers learn patterns from data instead of being explicitly programmed with rules for every possible situation. A machine learning model is trained using examples and can then make predictions or classifications on new data.

3. Types of Machine Learning

- **Supervised Learning:** The model learns from labeled examples. Classification and regression are common supervised learning tasks.
- **Unsupervised Learning:** The model works with data without target labels and attempts to discover useful patterns or groups.
- **Reinforcement Learning:** An agent learns by interacting with an environment and receiving rewards or penalties for its actions.

4. Common Machine Learning Algorithms

Linear Regression is commonly used for predicting continuous numerical values. Logistic Regression can be used for classification. Decision Trees make predictions using a sequence of rule-based decisions. K-Nearest Neighbors classifies an example according to nearby training examples. Support Vector Machines find a decision boundary that separates classes. K-Means is an unsupervised algorithm used for clustering.

5. Natural Language Processing

Natural Language Processing (NLP) is a field of AI focused on enabling computers to work with human language. NLP applications include sentiment analysis, text classification, machine translation, information extraction, question answering, and conversational assistants.

6. Chatbots

A chatbot is a software application that communicates with users through natural language. A chatbot can use predefined rules, machine learning models, retrieval systems, or generative AI models. A document-aware chatbot can retrieve relevant information from uploaded documents and use that information to answer user questions.

7. Document Question Answering

Document question answering allows a user to ask questions about the contents of a document. A typical system first extracts text from the document, divides the text into smaller sections, represents the sections in a searchable form, retrieves the sections most relevant to a question, and produces an answer from the retrieved information.

8. Important Concepts

- **Training data:** Data used to learn patterns and relationships.
- **Feature:** An input variable used by a machine learning model.
- **Label:** The target value associated with a training example in supervised learning.
- **Model:** A learned representation used to make predictions or decisions.
- **Inference:** Using a trained model to produce an output for new input.
- **Accuracy:** The proportion of predictions that are correct.

9. Quick Facts for Testing JARVIS

- AI is the broader field, while machine learning is one approach within AI.
- Supervised learning uses labeled training examples.
- K-Means is commonly used for clustering.
- NLP deals with human language and text.
- A document chatbot can retrieve relevant passages from an uploaded PDF.
- Inference means using a trained model with new input.

Suggested test question: What are the three types of machine learning mentioned in this document?

Expected answer: Supervised Learning, Unsupervised Learning, and Reinforcement Learning.